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Exploring the Effects of Prosocial and Self-Kindness Interventions on Mental Health Outcomes

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A growing body of literature suggests that prosocial behavior, or behavior intended to help others, benefits well-being. However, modern society often places a greater emphasis on self-focused acts in the pursuit of well-being. To understand the effects of these differing forms of kindness (to others or the self), we conducted a 2-week intervention study of a community sample during the COVID-19 pandemic in 2020. Participants were randomly assigned to a prosocial (other-kindness), self-kindness, or control condition, and those in the active conditions were asked to perform three acts of kindness each week. Of those who completed the intervention ($N = 777$), we found that participants in the other-kindness (vs. control) group experienced significant decreases in depression, anxiety, and loneliness from pre- to postintervention, offering compelling evidence for the mental health benefits of prosocial behavior. Furthermore, we found that participants in the self-kindness (vs. control) group experienced significant decreases in depression, but no differences in anxiety and loneliness. While the self-kindness group reported experiencing more positive feelings during their acts of kindness, the other-kindness group felt more connected. Exploratory mediation analyses revealed that, for the prosocial group, the effect of condition on depression, anxiety, and loneliness was mediated by increases in feelings of social connection, whereas for the self-kindness group, the effect of condition on depression was mediated by increases in positive feelings. Overall, these findings reaffirm the benefits of prosocial behavior on well-being and suggest unique pathways to mental health benefits for these two forms of kindness.

Keywords: prosocial behavior, self-kindness, mental health

The pursuit of well-being is a central aspect of the human experience. A burgeoning body of research illustrates that prosocial behavior, or behavior intended to benefit others, makes people happy (Aknin et al., 2013; Dunn et al., 2008) and is even linked with reduced mortality (Brown et al., 2003). Despite these findings, acts of kindness toward others are often portrayed as taxing. Phrases like “you can’t pour from an empty cup” suggest that giving to others depletes our own resources. However, another form of kindness—self-kindness—is often portrayed in the opposite manner. It is likely you have been told that you deserve to buy yourself a cup of coffee, read an article about the benefits of focusing on yourself, or seen an advertisement touting the personal transformation that will accompany a spa day (Carmel, 2022; Raypole, 2021). While undoubtedly pleasant, do these self-focused strategies promote well-being? If so, are they equally effective as doing acts of kindness for others? Prior work has compared these behaviors, revealing that prosociality has a greater influence on positive affect, psychological flourishing, and immune-related gene expression than self-focused acts of kindness (Nelson et al., 2016; Nelson-Coffey et al., 2017; Regan et al., 2022). However, research has also found that acts of self-kindness increase happiness just as well as acts of kindness to others (Rowland & Curry, 2018). The present study aimed to extend this work by directly comparing the effects of prosocial and self-kindness interventions, with a specific focus on how they influence negative mental health outcomes, such as depression, anxiety, and loneliness.

Prosocial behavior is defined as any action intended to help others, regardless of the benefit to the self (Batson & Powell, 2003). The positive effects of prosocial behavior are far-reaching. Experimental studies have shown that prosocial spending is associated with greater self-reported happiness compared to personal (i.e., self-focused) spending (Dunn et al., 2008), a finding echoed in cross-cultural research (Aknin et al., 2013). Correlational research has identified an association between greater self-reports of helping others and better mental health (Schwartz et al., 2003), as well as between daily acts of prosocial behavior and reduced everyday

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stress (Raposa et al., 2016). Importantly, prosocial behavior appears to contribute to these benefits for well-being; in an experimental study, it was found that prosocial behavior, operationalized as giving gifts to others or writing a note of appreciation for another, reduced loneliness more than a self-kindness or control condition (Lanser & Eisenberger, 2023). Beyond these psychological benefits, longitudinal research has underscored the long-term health benefits of volunteering, such as lower blood pressure and decreased mortality rates (Brown et al., 2003, 2009; Piliavin & Siegl, 2007; Poulin et al., 2013). Engaging in prosocial acts often spurs social interaction with strangers, which has been shown to boost positive experiences more than individuals anticipate (Epley & Schroeder, 2014; Schroeder et al., 2022). Thus, the social interaction that accompanies prosocial acts may confer additional benefits.

While the impact of prosocial behavior on positive well-being is well-established, the effects of self-directed acts of kindness are less clear. This may be because many behaviors that could fall into the category of self-kindness are studied as individual constructs, such as exercise, healthy eating, and meditation. While there is substantial research on these examples, it is difficult to determine whether positive effects are due to improved health or the act of engaging in self-kindness. The few studies that have directly compared the effects of prosocial and self-kindness behaviors are somewhat mixed. In an intervention comparing prosocial behavior to self-kindness, Nelson et al. (2016) found that prosocial behavior led to greater psychological flourishing (as measured by a composite self-report measure of emotional, psychological, and social well-being) than self-kindness, an effect mediated by increased positive emotions and decreased negative emotions. Similarly, a randomized control trial demonstrated that a prosocial condition, but not a self-kindness condition, led to reductions in a biomarker linked with stress-induced immune responses (conserved transcriptional response to adversity [CTRA] gene expression; Nelson-Coffey et al., 2017). Another intervention found that performing acts of kindness for the self, for strong social ties, for weak social ties, and observing acts of kindness all increased happiness, with no significant group differences (Rowland & Curry, 2018). Taken together, these findings suggest that both acts of kindness for the self and others likely feel pleasurable, but there may be unique benefits of prosocial acts that are not paralleled by self-focused acts of kindness. Moreover, these studies have not examined the effects of these acts on specific dimensions of mental health, such as depression and anxiety, and have relied on moderately sized samples.

The present study examined the effects of self-kindness and other-kindness on depression, anxiety, and loneliness through a 2-week kindness intervention in a large community sample during the COVID-19 pandemic. This investigation was particularly important during the COVID-19 pandemic, when many individuals were engaging in social distancing and spending more time alone. Participants were randomly assigned to a self-kindness, other-kindness, or control group. Those in the kindness conditions were asked to complete three acts of either self-kindness or other-kindness during each week of the intervention. Prior to the intervention and at the end of the intervention, all participants completed questionnaires to assess aspects of mental health, which we operationalized as depression, anxiety, and loneliness. Participants in the control condition completed these pre and postintervention measures but were not given task instructions during the intervention weeks. At the end of each week of the intervention, participants in the kindness conditions were asked to report on their acts of kindness, and all participants completed measures of affect and perceived stress about the past week.

Based on prior work illustrating the positive effects of prosocial behavior on well-being, we hypothesized that participants in the prosocial group would have significant decreases in depression, anxiety, and loneliness from pre- to postintervention and that there would be no changes for participants in the self-kindness and control groups. We further hypothesized that during the intervention, participants in both kindness conditions would find pleasure in their activities, leading to higher positive emotions, lower negative emotions, and lower stress than the control group. Last, we hypothesized that the other-kindness group would feel more connected during their activities.

Method

Participants

We aimed for a large community sample of 1,000 participants from the United States, recruited using the online recruitment platform Prolific. Of the 1,055 who began the first survey, 56 were excluded for incomplete responses. Of the 999 who completed the first survey, 777 completed the intervention and postintervention measures (43.4% female, $M_{\text{age}} = 31.7$, range = 18–57; 11.2% Asian, 7.0% Black, 6.1% mixed, 66.7% White, 2.2% other, 7.0% did not report). All study procedures were completed between October and December of 2020 during the COVID-19 pandemic, and 53.0% of participants were under stay-at-home restrictions. Participants varied in geographic location (32.2% large city, 32.4% suburbs of a large city, 22.1% small city, 7.5% town or village, and 5.8% rural area). At baseline, 57.4% of the sample had minimal or no depression (score of 0–10), 18.0% had mild depression (score of 11–16), 15.6% had moderate depression (score of 17–25), and 9.0% had severe depression (score of 26–60). Participants were not selected or screened for elevated mental health symptoms.

Participants were compensated a total of $25 for their participation in the study: $10 for completing the baseline survey (40 min) and $15 for completing the follow-up survey (45 min). Demographic details for each condition are reported in Table 1.

Procedure

Study procedures were approved by the University of California, Los Angeles (UCLA) Institutional Review Board (IRB No. 20-001098). Participants completed all surveys online, and informed consent was collected prior to administering the surveys. After completion of baseline measures to assess depression, anxiety, and loneliness, participants were randomly assigned to one of three experimental conditions: prosocial, self-kindness, or control. Participants in the prosocial group were instructed to complete three kind acts for others each week during the 2 weeks of the intervention. Specific instructions were as follows:

> During the course of our daily lives, we perform acts of kindness, generosity, and thoughtfulness—both large and small—for others. During these exceptional and socially distant times, these types of acts might include sending a text or making a phone call just to say hi, dropping off or sending a treat, link, or anything else that you know a friend or family member might enjoy, checking in on family members and friends, or writing a thank you letter. Over the next two weeks, *each week*, we would like you to perform *three* nice things for others during

Table 1
Demographic Information for Each Condition

Condition	% female	M_{age}	Ethnicity	% stay-at-home	Geographic area
Prosocial	45.8	31.2	14.8% Asian 6.0% Black 8.3% mixed 62.5% White 2.8% other 5.6% did not report	50.5	31.0% large city 34.7% suburbs of a large city 21.8% small city 8.8% town or village 3.7% rural area
Self-kindness	44.0	33.0	11.4% Asian 5.7% Black 5.7% mixed 66.3% White 2.8% other 8.2% did not report	53.2	34.4% large city 31.6% suburbs of a large city 20.9% small city 5.0% town or village 8.2% rural area
Control	40.9	30.9	8.2% Asian 9.0% Black 4.7% mixed 70.3% White 1.1% other 6.8% did not report	54.8	30.8% large city 31.5% suburbs of a large city 23.7% small city 9.0% town or village 5.0% rural area

> the course of the week. We would like for you to select the acts you perform and the times you perform them—these acts do not need to be for the same person, the person may or may not be aware of the act, and the act may or may not be similar to the ones listed above.

Participants in the self-kindness group were instructed to complete three kind acts for themselves each week of the intervention:

> During the course of our daily lives, we all perform acts of kindness for others, but we often neglect to do nice things for ourselves. Over the next two weeks, *each week*, we would like you to perform *three* nice things for yourself during the course of the week. During these exceptional times, these types of acts might include larger acts (e.g., enjoying a day trip to your favorite hiking spot or treating yourself to a favorite meal) or they could be small (e.g., taking a 5-minute break when feeling stressed or taking time to engage in your favorite hobby), but they should be something out of the ordinary that you do for yourself with a little extra effort. We would like for you to select the nice things to do and the times you do them—these nice things for yourself do not need to be the same as the examples listed above and they should be things that you do explicitly for yourself, not others.

Participants in the control condition were not given any specific task instructions. Participants in the prosocial and self-kindness groups were texted reminders throughout the week to complete their three acts of kindness. At the end of each week, participants received an email with instructions to complete an online questionnaire. Participants in the prosocial and self-kindness conditions were asked to report on their acts of kindness, and all participants were asked to report on emotions and stress during the week. At the end of the second week, all participants completed a follow-up survey, which included the baseline measures of depression, anxiety, and loneliness, in addition to the measures collected at the end of the first week of the intervention.

Measures

Baseline and Follow-Up Measures

Depression. The Beck Depression Inventory was used to assess depression (Beck et al., 1961, 1996) at baseline and postintervention. The Beck Depression Inventory is a 21-item questionnaire that measures symptoms of depression over the past week on a scale of 0–3. An example question includes the response options "I do not feel sad," "I feel sad," "I am sad all the time and can't snap out of it," and "I am so sad or unhappy that I can't stand it." One question regarding suicidality was excluded from the measure for our study, and the clinical cutoff values have been adjusted for this. We found high internal reliability of the measure in our sample (baseline Cronbach's $\alpha = .93$, follow-up Cronbach's $\alpha = .93$).

Anxiety. The state subscale of the State Trait Anxiety Index was used to assess current feelings of anxiety (Spielberger, 1983) at baseline and postintervention. The State Trait Anxiety Index asks participants to indicate their level of agreement with 20 items such as "I feel calm" and "I am tense" on a 4-point scale, ranging from *not at all* to *very much so*. We found high internal reliability for this measure in our sample (baseline Cronbach's $\alpha = .95$, follow-up Cronbach's $\alpha = .94$).

Loneliness. The UCLA Loneliness Scale v3 (UCLALS) was used to assess feelings of loneliness (Russell, 1996) at baseline and postintervention. The UCLALS asks participants to indicate how often they feel a certain way, such as "How often do you feel alone?" on a 4-point scale, ranging from *never* to *always*. The UCLALS includes 20 items and had high internal reliability in our sample (baseline Cronbach's $\alpha = .95$, follow-up Cronbach's $\alpha = .95$).

Intervention Measures

Reporting on Acts. Each week, participants in the prosocial and self-kindness groups (but not the control group) were emailed and asked to complete a questionnaire reporting on the three activities that they chose to engage in. These questions included an open-ended field about what each participant did and information about the activity they engaged in (i.e., how long the activity took, what day they did it, if they had ever done the activity before, and if so, how often). Examples of prosocial acts included calling friends and family who were going through difficult times, cooking and delivering food to others, and helping others with responsibilities such as childcare and grocery shopping. Examples of acts of self-kindness included taking baths, buying gifts for oneself, and setting aside time to relax with video games and television.

Participants were also asked to record how much they enjoyed the activity on a scale of 1 to 10 and how it made them feel (good, happy, satisfied, connected, and in a positive mood) on a 7-point scale from *not at all* to *extremely*. A mean score of "positive feelings" was created using the average scores for feeling good, happy, satisfied, and in a positive mood due to high correlations between these items ($rs > .85$). Ratings of positive feelings were highly correlated across intervention weeks among both prosocial, Week 1: M (SD) = 5.5 (1.1); Week 2: M (SD) = 5.4 (1.2); $r = .70$; and self-kindness conditions, Week 1: M (SD) = 5.9 (0.8); Week 2: M (SD) = 6.0 (0.8); $r = .56$. Ratings of social connection were also highly correlated across the intervention weeks among both prosocial, Week 1: M (SD) = 5.4 (1.3); Week 2: M (SD) = 5.4 (1.3); $r = .68$; and self-kindness conditions, Week 1: M (SD) = 4.7 (1.6); Week 2: M (SD) = 4.9 (1.6); $r = .70$. Because of this, average scores across weeks were used in analyses.

Emotions. An abbreviated version of the Positive and Negative Affect Schedule, completed by all participants, was used to measure positive and negative affect at the end of each week of the intervention (Watson et al., 1988). The measure asks participants to rate the extent to which they experienced nine emotions over the past week on a 7-point scale ranging from *not at all* to *extremely*. Positive items include happy, pleased, joyful, and fun/enjoyment (Week 1 Cronbach's $\alpha = .93$, Week 2 Cronbach's $\alpha = .93$). Negative items include angry/hostile, worried/anxious, depressed/blue, frustrated, and unhappy (Week 1 Cronbach's $\alpha = .91$, Week 2 Cronbach's $\alpha = .90$). Positive affect across intervention weeks were highly correlated for the prosocial, Week 1: M (SD) = 4.5 (1.3) and Week 2: M (SD) = 4.6 (1.3); $r = .78$; self-kindness, Week 1: M (SD) = 4.7 (1.3) and Week 2: M (SD) = 4.8 (1.3); $r = .76$; and control conditions, Week 1: M (SD) = 4.0 (1.5) and Week 2: M (SD) = 4.1 (1.5); $r = .79$, as were negative affect across intervention weeks were highly correlated for the prosocial, Week 1: M (SD) = 2.5 (1.3) and Week 2: M (SD) = 2.5 (1.2); $r = .76$; self-kindness, Week 1: M (SD) = 2.4 (1.3) and Week 2: M (SD) = 2.3 (1.2); $r = .74$; and control conditions, Week 1: M (SD) = 2.7 (1.4) and Week 2: M (SD) = 2.6 (1.3); $r = .83$. For this reason, average scores across the 2 weeks were used to compare scores between groups.

Perceived Stress. The Perceived Stress Scale, completed by all participants, was used to measure how stressed participants felt during each week of the intervention (Cohen et al., 1983). The measure asks participants to rate how often they felt a certain way during the past week on a 5-point scale from *never* to *very often*. Sample items include "How often have you been upset because of something that happened unexpectedly?" and "How often have you felt that you were unable to control the important things in your life?" The Perceived Stress Scale had high internal reliability in our sample (Week 1 Cronbach's $\alpha = .90$, Week 2 Cronbach's $\alpha = .91$). Perceived stress across intervention weeks were highly correlated for the prosocial, Week 1: M (SD) = 15.9 (7.6) and Week 2: M (SD) = 15.2 (7.6); $r = .82$; self-kindness, Week 1: M (SD) = 15.1 (7.8) and Week 2: M (SD) = 13.9 (7.9); $r = .81$; and control conditions, Week 1: M (SD) = 16.6 (8.1) and Week 2: M (SD) = 15.9 (8.1); $r = .83$. Average scores across the 2 weeks were used to compare scores between groups.

Data Analytic Strategy

Primary analyses were performed using Jamovi Version 2.5.7.0 (R Core Team, 2021; The Jamovi Project, 2023). Mixed effect linear models, with Satterthwaite method for degrees of freedom, were used to test the effect of Group (Prosocial vs. Self-Kindness vs. Control) × Time (Baseline vs. Postintervention) interactions on outcomes of depression, anxiety, and loneliness using the GAMLj linear analysis module (Gallucci, 2019). Group and time were modeled as fixed effects, and random intercepts were modeled at the level of the individual. To test whether baseline symptom severity moderated treatment outcomes, we used analysis of covariance with postintervention scores as the dependent variables, baseline scores as centered covariates, and Group × Baseline interaction terms. This approach was used to avoid collinearity between baseline values and repeated-measures outcomes. Exploratory moderator analyses were used to examine gender differences in the observed effects (by testing Group × Time × Gender interactions) as well as to examine if the effects were moderated by stay-at-home restrictions (by testing Group × Time × Stay-at-Home Restriction interactions). Analyses of variance (ANOVAs) were used to compare the effect of condition on affect and perceived stress, averaged across the 2 weeks of the intervention. Independent sample t-tests were used to compare positive feelings and connection averaged across the 2-week intervention period in the prosocial and self-kindness groups. Last, exploratory mediation analyses were used to test whether positive feelings and connection mediated the effect of condition on depression and anxiety using Model 4 of the PROCESS macro and 5,000 bootstrap samples in SPSS (Hayes, 2022). Analyses were conducted with postintervention scores of depression, anxiety, and loneliness as the outcome variable and average social connection and positive feeling scores as the mediator. Preintervention scores of depression, anxiety, and loneliness were included as covariates for each respective mediation analysis.

Transparency and Openness

This study was not preregistered. All study materials, data, and analysis scripts are available on the Open Science Framework at https://osf.io/jvbcn/?view_only=b8b85cc5f92b47d38743846104c3b99e (Naclerio & Eisenberger, 2025). We aimed for a large national sample of 1,000 participants. Of the 1,055 who began the first survey, 56 were excluded for incomplete responses. Of the 999 who completed the first survey, 777 completed the intervention and postintervention measures. We report on all procedures and measures in the study.

Results

Baseline Comparisons

In examining baseline differences between our groups for participants who completed the entire study ($N = 777$), we found no significant differences in baseline measures of depression, Beck Depression Inventory: $F(2, 774) = 0.43$, $p = .649$; anxiety, State

Table 2

Means, Standard Deviations, and Correlations Between Mental Health Measures at Baseline for Those Who Completed the Study

Measure	*M*	*SD*	1	2	3
1. Beck Depression Inventory	11.0	10.0	—		
2. State Trait Anxiety Index	40.7	12.8	.74*** [.71, .77]	—	
3. University of California, Los Angeles Loneliness Scale	44.6	12.6	.62*** [.58, .66]	.60*** [.55, .64]	—

Note. $N = 777$. Values in brackets indicate the 95% confidence interval for each correlation.
*** $p < .001$.

Trait Anxiety Index: $F(2, 774) = 0.47$, $p = .627$; or loneliness, UCLALS: $F(2, 774) = 2.82$, $p = .060$.

Correlations between baseline measures of depression, anxiety, and loneliness can be found in Table 2. Although depression and anxiety were highly correlated, we kept them as separate measures because they are still well-known to assess different constructs.

Effects of Group Assignment on Depression, Anxiety, and Loneliness Over the Course of the Intervention

Depression

The model revealed a significant interaction between time and group on depression, $F(2, 774) = 4.97$, $p = .007$ (Figure 1A). Interaction contrasts revealed the prosocial group had greater reductions in depression compared to the control group, $b = -1.24$, 95% confidence interval (CI) [−2.22, −0.27], $t(774) = -2.50$, $p = .013$, $\beta = -0.13$, and that the self-kindness group had greater reductions in depression compared to the control group, $b = -1.33$, 95% CI [−2.23, −0.42], $t(774) = -2.87$, $p = .004$, $\beta = -0.13$. There were no significant differences in change in depression between the prosocial and self-kindness conditions, $b = 0.08$, 95% CI [−0.89, 1.05], $t(774) = 0.17$, $p = .867$, $\beta = 0.01$. Analyses of the simple effects revealed that participants in all groups showed significant decreases in depression from baseline to postintervention, prosocial group: $b = -1.95$, 95% CI [−2.68, −1.22], $t(774) = -5.23$, $p < .001$, $\beta = -0.20$; self-kindness group: $b = -2.03$, 95% CI [−2.67, −1.39], $t(774) = -6.23$, $p < .001$, $\beta = -0.20$; control group: $b = -0.71$, 95% CI [−1.35, −0.06], $t(774) = -2.15$, $p = .032$, $\beta = -0.07$.

Anxiety

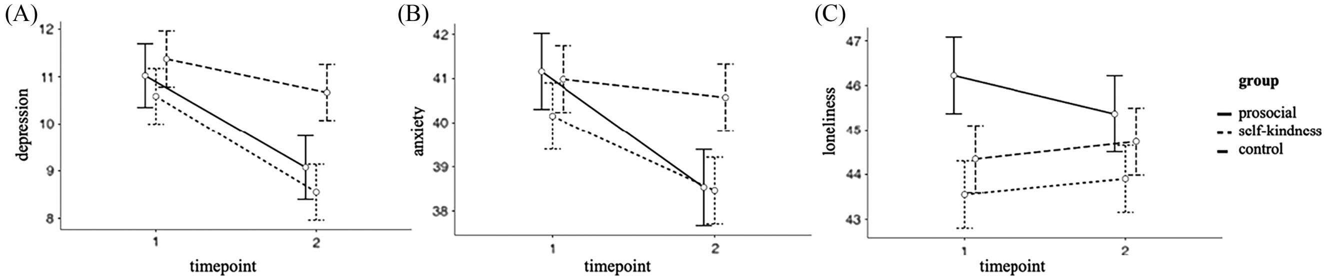
The model revealed a significant interaction between time and group on anxiety, $F(2, 773) = 4.37$, $p = .013$ (Figure 1B). Interaction contrasts revealed the prosocial group had greater reductions in anxiety compared to the control group, $b = -2.21$, 95% CI [−3.70, −0.72], $t(773) = -2.92$, $p = .004$, $\beta = -0.18$, but that the self-kindness group did not differ significantly from the control group, $b = -1.28$, 95% CI [−2.67, 0.11], $t(773) = -1.80$, $p = .072$, $\beta = -0.10$. There were no significant differences in change in anxiety between the prosocial and self-kindness conditions, $b = -0.94$, 95% CI [−2.43, 0.55], $t(773) = -1.24$, $p = .216$, $\beta = -0.07$. Analyses of simple effects revealed that participants in the prosocial and self-kindness groups experienced a significant decrease in anxiety from baseline to postintervention, prosocial group: $b = -2.63$, 95% CI [−3.75, −1.51], $t(773) = -4.61$, $p < .001$, $\beta = -0.21$; self-kindness group: $b = -1.69$, 95% CI [−2.67, −0.71], $t(774) = -3.38$, $p < .001$, $\beta = -0.13$, whereas the control group did not, $b = -0.42$, 95% CI [−1.40, 0.57], $t(773) = -0.83$, $p = .407$, $\beta = -0.03$.

Loneliness

The model revealed a significant interaction between time and group on loneliness, $F(2, 774) = 3.35$, $p = .036$ (Figure 1C). Interaction contrasts revealed the prosocial group had greater reductions in loneliness compared to the control group, $b = -1.25$, 95% CI [−2.31, −0.19], $t(774) = -2.32$, $p = .020$, $\beta = -0.10$ and compared to the self-kindness group, $b = -1.21$, 95% CI [−2.27, −0.16], $t(774) = -2.25$, $p = .024$, $\beta = -0.10$. There were no

Figure 1

Scores at Baseline (Timepoint 1) and Postintervention (Timepoint 2) in the Prosocial, Self-Kindness, and Control Groups for (A) Depression, (B) Anxiety, and (C) Loneliness



Note. Bars represent standard error.

significant differences in change in loneliness between the self-kindness and control conditions, $b = -0.04$, 95% CI [−1.02, 0.95], $t(774) = -0.08$, $p = .937$, $\beta = -0.00$. Analyses of the simple effects revealed that participants in the prosocial group experienced a significant decrease in loneliness from baseline to postintervention, $b = -0.86$, 95% CI [−1.66, −0.07], $t(774) = -2.13$, $p = .034$, $\beta = -0.07$, but there was no significant change in loneliness for the self-kindness group, $b = 0.35$, 95% CI [−0.34, 1.05], $t(774) = 0.99$, $p = .322$, $\beta = 0.03$ or the control group, $b = 0.39$, 95% CI [−0.31, 1.09], $t(774) = 1.10$, $p = .273$, $\beta = 0.03$.

Moderation Analyses With Baseline Symptom Severity

Analysis of covariance models testing Group × Baseline interactions on postintervention outcomes showed no significant moderation effects for depression, $F(2, 771) = 1.81$, $p = .165$; anxiety, $F(2, 770) = 0.09$, $p = .918$; or loneliness, $F(2, 771) = 0.74$, $p = .479$, suggesting that baseline symptom severity did not moderate the effects of the intervention.

Moderation Analyses With Gender and Stay-at-Home Restriction

There were no significant Group × Time × Gender interactions on depression, $F(4, 768) = 1.37$, $p = .244$; anxiety, $F(4, 767) = 0.04$, $p = .996$; or loneliness, $F(4, 768) = 0.62$, $p = .650$, suggesting that gender did not have a significant influence on the changes in depression, anxiety, or loneliness that were observed across conditions. There were also no significant Group × Time × Stay-at-Home interactions on depression, $F(2, 771) = 0.20$, $p = .827$; anxiety, $F(2, 770) = 0.64$, $p = .528$; or loneliness, $F(2, 771) = 0.47$, $p = .624$.

Emotions and Perceived Stress Across Weeks 1 and 2

To better understand the affective consequences of these interventions, we next examined the effect of the intervention on average ratings of positive affect, negative affect, and stress during the 2 weeks that the intervention ensued.

Positive Affect

A Levene's test revealed a violation of the assumption of homogeneity, $F(2, 774) = 5.19$, $p = .006$, so one-way Welch ANOVA was conducted. As expected, analyses revealed a significant difference in positive affect between groups, $F(2, 501) = 20.82$, $p < .001$ (Figure 2A). Post hoc pairwise comparisons using the Games-Howell test revealed that average positive affect scores over the 2-week intervention were significantly higher in the prosocial group compared with the control group, $t(486) = 4.48$, $p < .001$; and in the self-kindness group compared with the control group, $t(546) = 6.33$, $p < .001$. There were no significant differences between the prosocial and self-kindness groups, $t(461) = -1.55$, $p = .267$.

Negative Affect

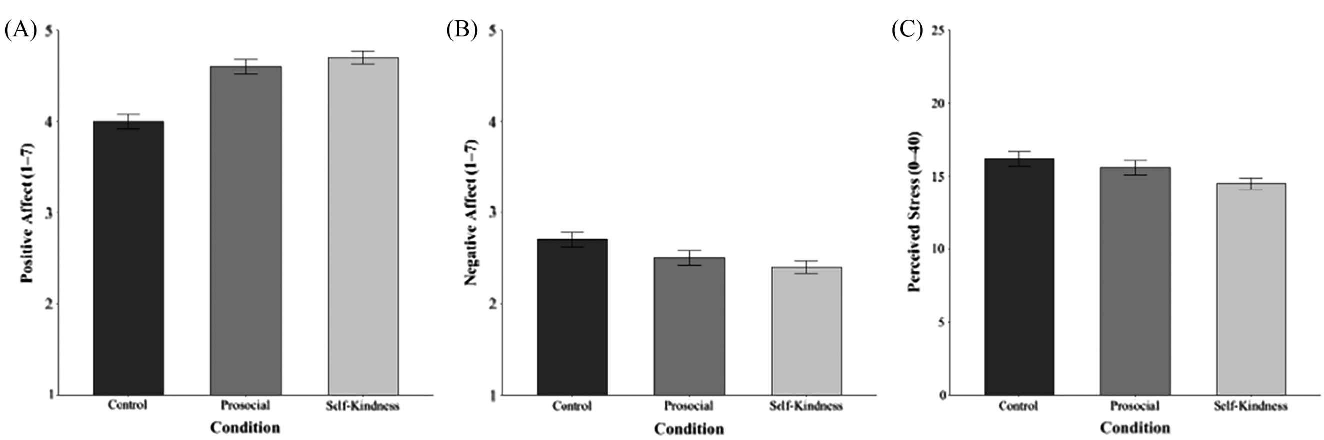
A Levene's test revealed a violation of the assumption of homogeneity, $F(2, 774) = 3.21$, $p = .041$, so one-way Welch ANOVA was conducted, which revealed a significant difference in average negative affect between groups, $F(2, 501) = 4.06$, $p = .018$ (Figure 2B). Pairwise comparisons using Games-Howell revealed that average negative affect scores over the 2-week intervention were significantly lower in the self-kindness group compared with the control group, $t(550) = -2.85$, $p = .013$. There were no significant differences between the prosocial group and the control group, $t(483) = -1.55$, $p = .271$; or between the prosocial and self-kindness groups, $t(461) = 1.19$, $p = .460$.

Perceived Stress

A one-way ANOVA showed that average perceived stress differed significantly between groups, $F(2, 774) = 3.78$, $p = .023$, $\omega^2 = .007$ (Figure 2C). Pairwise comparisons using Tukey's Honestly Significant Difference test revealed that average perceived stress scores over the 2-week intervention were significantly lower in the self-kindness group compared with the control group, $t(774) = -2.73$, $p = .018$, $d = -0.23$. There were no significant differences between the prosocial group and the control group, $t(774) = -0.97$, $p = .595$, $d = -0.09$; or between the prosocial and self-kindness groups, $t(774) = 1.57$, $p = .258$, $d = 0.14$.

Figure 2

Mean Scores Across Intervention Weeks in the Prosocial, Self-Kindness, and Control Groups for (A) Positive Affect, (B) Negative Affect, and (C) Perceived Stress



Note. Bars represent standard error.

In sum, we found that participants in the prosocial group experienced higher positive affect compared with the control group, but there were no significant differences in negative affect or stress. Meanwhile, participants in the self-kindness group experienced higher positive affect, lower negative affect, and lower stress over the 2-week intervention when compared with the control group. There were no significant differences in affect or stress between the self-kindness and prosocial groups.

Feelings During Acts of Kindness

To better understand the different consequences of the active intervention groups (prosocial vs. self-kindness), we examined positive feelings and connection while participants were engaging in acts of kindness. Welch's *t*-tests were run to determine whether there were differences between active intervention groups on positive feelings and connection due to the assumption of homogeneity of variances being violated, for positive feelings: $F(1, 493) = 24.45$, $p < .001$; for connection: $F(1, 495) = 14.45$, $p < .001$. While participants in the self-kindness group reported having more positive feelings, $t(351) = 6.30$, $p < .001$, $d = 0.59$, 95% CI [0.42, 0.78], participants in the prosocial group reported significantly more connection during their acts compared with the self-kindness group, $t(494) = 4.98$, $p < .001$, $d = 0.44$, 95% CI [0.26, 0.62]. Thus, while self-kindness may elicit more positive feelings, prosocial behavior is associated with greater feelings of social connection in the moment.

Exploratory Mediation Analyses

Based on our finding that the prosocial group reported significantly more social connection compared to the self-kindness condition, while the self-kindness group experienced more positive feelings, we conducted exploratory mediation analyses to test whether the effects of condition on change in mental health symptoms may occur through different mechanisms for prosocial versus self-kindness. The control condition did not complete ratings of social connection or positive feelings, so they were excluded from these analyses. To investigate the underlying mechanism by which prosocial behavior reduces depression, anxiety, and loneliness, we conducted mediation analyses with social connection as the mediator. To investigate the underlying mechanism by which self-kindness behavior reduces depression, we conducted a mediation analysis with the composite positive feeling measure as the mediator. We did not have baseline measures of social connection or positive feelings, and these variables were not collected for the control group, so we note that this approach is limited and exploratory.

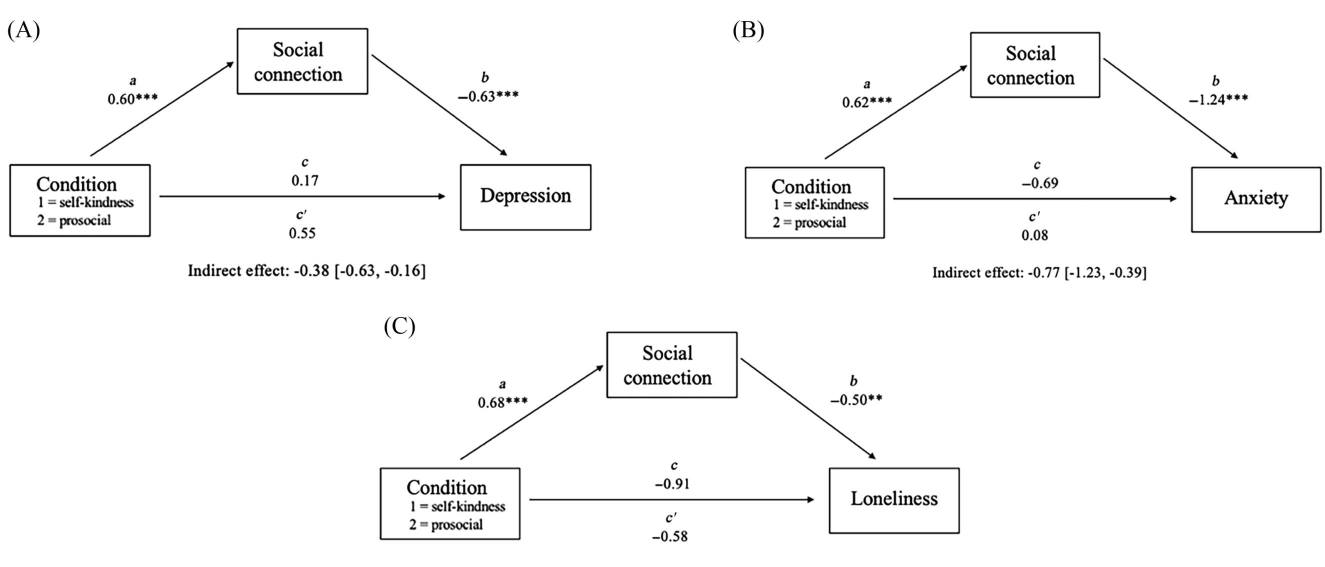
Social Connection as a Mediator of the Effect of Prosocial Behavior on Depression, Anxiety, and Loneliness

For depression, the overall model, including condition, average social connection, and preintervention depression as a covariate explained a significant portion of the variance in postintervention depression, $R^2 = .713$, $F(3, 493) = 407.25$, $p < .001$. The direct effect of the group was not significant, $b = 0.55$, $t(494) = 1.21$, $p = .227$. However, the indirect effect was significant ($b = -0.38$, 95% CI [−0.63, −0.16]), suggesting the increases in social connection among the prosocial group mediated the effect of condition on depression (Figure 3A).

For anxiety, the overall model explained a significant portion of the variance in postintervention anxiety, $R^2 = .615$, $F(3, 492) = 261.86$, $p < .001$. The direct effect of the group was not significant,

Figure 3

Mediation Models Showing the Indirect Effect of Condition (Prosocial vs. Self-Kindness) on (A) Depression, (B) Anxiety, and (C) Loneliness Through Social Connection



Note. All models include baseline scores of outcome variables as covariates. Coefficients are unstandardized, and 95% confidence intervals are shown in brackets.
** $p < .01$. *** $p < .001$.

$b = 0.08$, $t(493) = 0.11$, $p = .909$. However, the indirect effect was significant ($b = -0.77$, 95% CI [−1.23, −0.39]), suggesting the increases in social connection among the prosocial group mediated the effect of condition on anxiety (Figure 3B).

For loneliness, the overall model explained a significant portion of the variance in postintervention loneliness, $R^2 = .795$, $F(3, 493) = 636.65$, $p < .001$. The direct effect of the group was not significant, $b = -0.58$, $t(494) = -1.06$, $p = .290$. However, the indirect effect was significant ($b = -0.34$, 95% CI [−0.65, −0.07]), suggesting the increases in social connection among the prosocial group mediated the effect of condition on loneliness (Figure 3C).

These analyses illustrate a potential pathway by which prosocial behavior reduces feelings of depression, anxiety, and loneliness through improvements in social connection.

Positive Feelings as a Mediator of the Effect of Self-Kindness on Depression

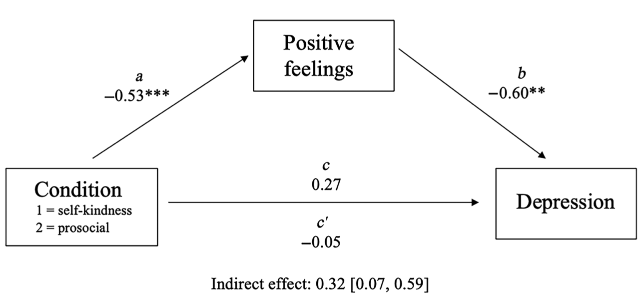
Given the significant reduction in depression, but not anxiety and loneliness, observed in the self-kindness group, we focused our analyses here on depression as the outcome. The overall model with condition, average positive feelings, and preintervention depression as a covariate explained a significant portion of the variance in postintervention depression $R^2 = .714$, $F(3, 491) = 407.73$, $p < .001$. The direct effect of the group was not significant, $b = -0.05$, $t(492) = -0.11$, $p = .915$. However, the indirect effect was significant ($b = 0.32$, 95% CI [0.07, 0.59]), suggesting the increases in positive feelings among the self-kindness group mediated the effect of condition on depression (Figure 4).

Discussion

The present study examined the effects of a prosocial versus self-kindness intervention on loneliness, anxiety, and depression in a community sample during the COVID-19 pandemic. As predicted, participants in the prosocial group experienced significant decreases in depression, anxiety, and loneliness compared with participants in the control group over the course of the 2-week intervention. We also found that participants in the self-kindness group experienced significant decreases in depression compared to the control group,

Figure 4

Mediation Model Showing the Indirect Effect of Condition (Prosocial vs. Self-Kindness) on Depression Through Positive Feelings



Note. The model includes baseline depression as a covariate. Coefficients are unstandardized, and 95% confidence intervals are shown in brackets.
$^{**}p < .01$. $^{***}p < .001$.

but there were no differences in anxiety or loneliness. Participants in both kindness conditions reported improved positive affect compared to the control group, but only the self-kindness group reported less negative affect and stress compared to the control group. No significant differences in affect or perceived stress were found between the prosocial and self-kindness groups. Participants in the prosocial condition also felt more connected during the intervention, while those in the self-kindness condition reported more positive feelings in response to engaging in the acts. For the prosocial group, improvements in social connection mediated the effect of condition on depression, anxiety, and loneliness. For the self-kindness condition, increases in positive feelings mediated the effect of condition on depression, suggesting dual routes to well-being for these two forms of kindness. These findings add to a growing body of research illustrating that prosocial behavior positively influences well-being, above and beyond acts of self-kindness (Nelson et al., 2016; Nelson-Coffey et al., 2017; Regan et al., 2022).

The improvements in mental health observed in the prosocial condition align with the growing consensus that helping others can foster better psychological health (Lanser & Eisenberger, 2023; Morelli et al., 2015). However, the underlying mechanism by which prosocial behavior improves well-being is still up for debate. Unlike self-kindness, which focuses on the self, prosocial behavior has the potential to shift focus away from the self and toward others. Indeed, a self-distanced perspective has been shown to improve emotional processing, buffer against depressed affect, and reduce cardiovascular reactivity (Ayduk & Kross, 2008; Kross & Ayduk, 2008; Kross et al., 2005). This effect can even be seen in language. Linguistic strategies, such as referring to oneself by name rather than with first-person pronouns, have been shown to improve emotion regulation and are associated with better mental health outcomes (Nook et al., 2022, 2025; Orvell et al., 2021). This shift in focus may also foster a sense of connection or belonging with others, which is critical for mental health (Baumeister & Leary, 1995; Holt-Lunstad et al., 2010). This aligns with our finding that the prosocial group felt more socially connected during the intervention than the self-kindness group, as well as our exploratory mediation analyses, which found that increases in social connection mediated the reductions in depression, anxiety, and loneliness observed in the prosocial group. However, we are unable to distinguish whether the benefits observed are solely due to the prosocial group interacting with others more (and thus prosocial behavior is one amongst many routes toward increased social connection) or whether prosocial behavior has unique benefits beyond increased social connection. Future research may include a condition that encourages nonprosocial social interactions to test this question. Another possibility is that those in the prosocial condition were more likely to receive kindness from those they helped in return for their behavior, which may have increased their well-being.

Interestingly, there were no significant differences in negative affect or perceived stress between the prosocial and control groups, suggesting that prosocial behavior is a complex process that may come with different affective experiences over time. Anticipating and even acting to help somebody might feel stressful before and during the act, while the gratitude received from helping and the warm glow that follows its completion might feel more positive. Across multiple experiments, Kumar and Epley (2023) found that givers underestimate the positive impact a prosocial act will have on a recipient, focusing more on their competence to complete the act rather than the warmth a recipient will feel. A giver's concern about competence may help explain why we did not observe reduced negative affect or perceived stress in the prosocial group.

While the prosocial group experienced improvements in depression, anxiety, and loneliness, the benefits of self-kindness were limited to depression. The improvements in depression align with our findings that the self-kindness group experienced boosts in positive affect and positive feelings, both of which can be deficient in depression (e.g., depressed mood, anhedonia). Indeed, our mediation analyses illustrated that increases in positive feelings mediated the effects of self-kindness on depression. Acts of self-kindness likely feel pleasurable, and these positive feelings may elicit reductions in depression. Although these benefits did not extend to anxiety or loneliness, we interpret this pattern cautiously. Depression and anxiety are highly correlated (Clark & Watson, 1991) and are often conceptualized as part of a broader internalizing dimension (Kotov et al., 2017), making differences difficult to interpret. Prior experimental studies comparing acts of self-kindness with prosocial and control conditions have found less favorable results regarding the benefits of self-kindness. One such study found no changes in positive affect, negative affect, or psychological flourishing in their self-kindness group, while another set of studies found no beneficial effects of engaging in self-kindness on Conserved Transcriptional Response to Adversity gene expression (Nelson et al., 2016; Nelson-Coffey et al., 2017; Regan et al., 2022). Additional work is needed to determine whether acts of self-kindness do improve well-being, and whether these benefits depend on the type, context, or timing of the act.

Given the importance of mental health and the popularity of "self-care," future research is necessary to determine how these various forms of kindness (to others vs. the self) influence well-being. While most acts of self-kindness likely feel good in the moment, some forms, such as exercise or healthy eating, may be more helpful in the long term than others, such as watching TV or eating desserts. The benefits of these acts may be limited by the duration of their effects on affect, limiting their potential for sustained improvements in well-being. Prosocial behavior, by contrast, may improve well-being through increasing feelings of social connection, which may have more enduring positive effects. Last, future work might examine strategies to increase engagement in prosocial behavior and thus improve the associated mental health outcomes.

Constraints on Generality and Limitations

The context of COVID-19 was both a strength and a limitation of this study. On the one hand, our findings offer insight into how acts of prosociality and self-kindness impact depression, anxiety, and loneliness during challenging times. It is encouraging that, even while much of the nation was on lockdown, engaging in acts to help others for only 2 weeks had the power to improve mental health. Our participants found ways to help others—often remotely or while social distancing—and these acts fostered a sense of connection. While previous research has illustrated that some benefits of prosocial behavior are greater than those of self-kindness (Nelson et al., 2016; Nelson-Coffey et al., 2017; Regan et al., 2022), this was the first study, to our knowledge, to investigate how these acts of kindness influence specific dimensions of mental health. Future work is necessary to determine whether this effect persists outside of the context of the COVID-19 pandemic, a time during which adults experienced an increased prevalence of mental health adversities and, specifically, an increase in depression (Czeisler et al., 2020; Ettman et al., 2020). A prosocial intervention that encouraged participants to find creative ways to help others may have been uniquely beneficial. Similarly, a self-kindness intervention that encouraged participants to step away from others and focus on themselves may have been less beneficial than typical when many were already spending much of their time alone.

We note that the instructions for the self-kindness condition mention that people often neglect to take care of themselves, while the instructions for the prosocial condition do not make the same comparison (such as mentioning that people often neglect to take care of others). We included this language in the self-kindness condition to encourage our participants to engage in the intervention despite the pressing concerns of COVID-19, but we acknowledge that this may have inadvertently created a sense of competing obligations in one condition but not the other. A future study should include parallel language in both conditions. Additionally, most of our participants resided in urban locations, which may have influenced the types and targets of the acts of kindness that were chosen. Future work may overcome this limitation by seeking participants from more rural locations as well. Other limitations of our study include the online data collection and reliance on self-report measures.

Future work might overcome these limitations by employing a multimodal approach, including clinical interviews to assess mental health symptomology or ecological momentary assessment to elucidate the time-course effects of prosocial and self-kindness behaviors. The findings of this study also have therapeutic implications. Acts of kindness toward others are emphasized in some forms of therapy, such as dialectical behavior therapy, in which contributing to others is taught as a strategy for managing distressing emotions (Linehan, 2014). Future research might examine how prosocial interventions influence depression and anxiety in therapeutic contexts and among clinical samples. Last, this study was conducted on an American sample, in part because of the prevalence of self-care practices in the United States. Further exploration should test whether the same effects would be observed in cultures that have a more interdependent self-construal and where caring for others plays a central role in daily life (Markus & Kitayama, 1991).

Conclusion

In conclusion, this study's comparison of prosocial and self-kindness interventions on mental health advances the literature on the relationship between prosocial behavior and well-being. Our findings demonstrate that prosocial behavior has the potential to reduce symptoms of depression, anxiety, and loneliness, while self-kindness has the potential to reduce symptoms of depression. Furthermore, exploratory analyses suggest that these effects may be driven by different mechanisms, such as social connection for prosocial behavior and positive feelings for self-kindness. Ultimately, our study underscores a compelling notion: in striving for personal well-being, helping others might be a powerful pathway to help ourselves.

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